

Sub-Tissue Radiation Response Analysis

Beyond Organs: 33 Sub-Tissues, Response Fingerprints & Radiation-Specific Patterns

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What This Analysis Adds

The previous analysis tested 5 broad organs (Root, Seedling, Leaf_Shoot, Flower, Seed_Silique). This analysis goes deeper in three ways: (1) tests 33 individual sub-tissues for radiation response enrichment, (2) computes a tissue-weighted response score that estimates how much each sub-tissue's transcriptome shifted - even though all experiments used whole seedlings, and (3) builds a radiation response fingerprint for each sub-tissue and tests whether tissue response profiles are radiation-type-specific or conserved across radiation conditions.

1. Sub-Tissue Level Enrichment Tests (33 Tissues)

The original analysis tested 5 broad organs. Now we test 29 sub-tissues (those with ≥ 50 tissue-specific genes) for differential DEG rates. The chi-square test is highly significant in all 4 testable comparisons:

Comparison	Chi-sq	df	p-value	Significant sub-tissues
radiation_effect (Co-60 100Gy)	546.1	28	2.1e-97	9 significant sub-tissues
Cs137_100cGy	162.4	28	6.8e-21	4 significant sub-tissues
Cs137_10cGy	145.3	28	8.7e-18	1 significant sub-tissue
genotype_interaction	51.5	28	0.004	0 after Bonferroni

Key sub-tissue findings (Co-60 100 Gy, Fisher's exact with Bonferroni):

Sub-tissue	Organ	Genes	DEGs	DEG rate	OR	padj	Result
senescing_leaf	Leaf_Shoot	840	118	14.1%	4.00	2.0e-27	enriched
cotyledon	Seedling	349	67	19.2%	5.40	2.1e-22	enriched
seedling_green_parts	Seedling	123	26	21.1%	5.68	1.9e-9	enriched
sepal	Flower	285	36	12.6%	3.08	1.5e-6	enriched
root	Root	1207	26	2.2%	0.42	6.6e-5	depleted
rosette_vegetative	Leaf_Shoot	219	0	0.0%	0.00	1.7e-3	depleted
seed_stage_6	Seed_Silique	256	1	0.4%	0.08	3.5e-3	depleted
stem	Leaf_Shoot	429	7	1.6%	0.33	2.8e-2	depleted
mature_pollen	Flower	2242	77	3.4%	0.68	4.1e-2	depleted

New finding invisible at organ level

At the broad organ level, Leaf_Shoot was enriched (OR=1.94). At the sub-tissue level, this enrichment is driven entirely by senescing_leaf (OR=4.00, 14.1% DEG rate) and cauline_leaf - not by rosette_leaf or shoot apices, which are depleted or neutral. Similarly, the Seedling enrichment is driven by cotyledon (19.2% DEG rate) and seedling_green_parts (21.1%), not by whole_plant_pre_bolting. The organ-level analysis averages away these dramatic differences.

2. Tissue-Weighted Response Scores

Since all experiments used whole seedlings, we cannot directly measure per-tissue expression changes. But we can estimate them. For each gene, we know what fraction of its expression comes from each sub-tissue (from the AtGenExpress atlas). Multiplying each gene's log2FC by its expression fraction in each tissue, then summing across all genes, gives a tissue-weighted response score - an estimate of how much each sub-tissue's transcriptome shifted.

Top and bottom sub-tissues by weighted response score (Co-60 100 Gy):

Sub-tissue	Organ	Score	Interpretation
senescing_leaf	Leaf_Shoot	0.225	Highest - senescence genes strongly upregulated
sepal	Flower	0.177	Sepal-specific genes strongly affected
cotyledon	Seedling	0.147	Cotyledon genes strongly upregulated
cauline_leaf	Leaf_Shoot	0.136	Cauline leaf genes affected
seedling_green_parts	Seedling	0.114	Seedling photosynthetic genes
...	...	...	...
rosette_vegetative	Leaf_Shoot	0.006	Nearly no response
inflorescence_apex	Leaf_Shoot	0.003	Minimal response
shoot_apex_transition	Leaf_Shoot	-0.004	Slightly negative (downregulated)

Permutation test validation:

To confirm these scores are not random, we permuted log2FC values across genes (keeping tissue expression fractions fixed) and recomputed the variance of tissue scores 1,000 times. In all 4 comparisons, the observed variance was hundreds of standard deviations above the permutation null:

Comparison	Observed var	Perm mean var	Z-score	p-value
radiation_effect	0.00258	0.000011	523.8	< 0.001
Cs137_100cGy	0.00203	0.000006	721.8	< 0.001
Cs137_10cGy	0.00119	0.000005	502.4	< 0.001
genotype_interaction	0.00371	0.000015	510.2	< 0.001

Interpretation

The tissue-weighted response scores are real, not artifacts of expression patterns. The permutation test shuffles which genes are up/down, keeping the tissue expression pattern fixed. If the observed variance is this much higher than random, it means the radiation response is genuinely non-uniform across tissues - some tissues' genes are systematically more affected than others.

3. Tissue-Specific Genes Respond More Strongly

A fundamental question: is the radiation response itself tissue-specific? If tissue-specific genes (high Tau) respond more strongly than constitutive genes (low Tau), then the radiation response is concentrated in specific tissues rather than uniformly distributed.

Comparison	Spearman rho	Wilcoxon p	DEG rate (Tau 0.2-0.4)	DEG rate (Tau 0.8-1.0)
radiation_effect	0.364	p ~ 0	0.2% (low Tau)	6.7% (high Tau)
Cs137_100cGy	0.412	p ~ 0	0.0%	2.8%
Cs137_10cGy	0.384	p ~ 0	0.0%	1.0%
genotype_interaction	0.381	p ~ 0	0.8%	3.3%

Key insight

The Spearman correlation between Tau and $|\log_2FC|$ is 0.36-0.41 across all comparisons. This means tissue-specific genes have systematically larger radiation responses. The DEG rate for the most tissue-specific genes (Tau > 0.8) is 10-25x higher than for constitutive genes (Tau 0.2-0.4). This is strong evidence that the radiation response is itself tissue-specific - it's not just that some tissues happen to have more DEGs, but that tissue-specific genes are intrinsically more responsive to radiation.

4. Are Tissue Responses Conserved Across Radiation Types?

If tissue-specific responses are a fundamental property of plant radiation biology, the tissue response ranking should be consistent across different radiation types. We tested this with Kendall's coefficient of concordance (W), which measures whether multiple raters (radiation types) agree on the ranking of items (sub-tissues).

Result: Kendall's W = 0.905 (p < 0.001)

The three gamma radiation types (Co-60 100 Gy, Cs-137 100 cGy, Cs-137 10 cGy) agree strongly on which sub-tissues respond most. The tissue response ranking is conserved across radiation types. However, the magnitude differs: Co-60 produces larger responses in photosynthetic tissues (senescing_leaf, cotyledon, cauline_leaf), while Cs-137 produces larger responses in root and seed tissues. The pairwise Spearman correlation between Cs-137 100 cGy and 10 cGy is 0.99 (nearly identical tissue ranking), while Co-60 vs Cs-137 is 0.80 (same general pattern but with radiation-type-specific differences).

Most consistent sub-tissues (low CV across radiation types):

Sub-tissue	Organ	CV	Interpretation
seed_stage_9	Seed_Silique	0.056	Seed development - conserved response
stamen	Flower	0.058	Male reproductive tissue - conserved
mature_pollen	Flower	0.062	Pollen genes - conserved response
root	Root	0.071	Root genes - conserved (but Cs-137 > Co-60)

Most radiation-type-specific sub-tissues (high CV):

Sub-tissue	Organ	CV	Interpretation
cauline_leaf	Leaf_Shoot	0.287	Co-60 drives much larger response
senescing_leaf	Leaf_Shoot	0.280	Co-60 drives much larger response
cotyledon	Seedling	0.273	Co-60 drives much larger response
whole_plant_pre_bolting	Seedling	0.267	Co-60 drives much larger response

5. Sub-Tissue Response Clusters

Hierarchical clustering of sub-tissues by their radiation response fingerprint (magnitude + direction across 4 comparisons) reveals 4 distinct response groups:

Cluster	Members	Response pattern
1: High-response photosynthetic	senescing_leaf, cotyledon, seedling_green_parts, cauline_leaf, sepal, flower_stage_15	Highest response magnitude, strongly upregulated by Co-60, moderately up by Cs-137, reversed by sog1-1
2: Moderate reproductive	stamen, petal, silique stages 3-5, seed_stage_6-7, hypocotyl, stem, rosette_leaf, whole_plant	Moderate response, mixed direction, seed/silique tissues shift from up (radiation) to down (interaction)
3: Low-response meristematic	shoot apices, inflorescence_apex, rosette_vegetative, flower stages 9-12, carpel, pedicel	Lowest response magnitude, near-zero direction, meristematic and early floral tissues barely respond
4: Root-seed cluster	root, mature_pollen, seed_stage_8-10	Root and late seed tissues cluster together - strong Cs-137 response, strong sog1-1 interaction

Biological significance

Cluster 4 is the most interesting: root and late-stage seed tissues (seed_stage_8-10) cluster together despite being in different broad organs. They share a response pattern that is disproportionately strong under Cs-137 (not Co-60) and strongly reversed by sog1-1. This suggests a shared SOG1-dependent DNA damage response pathway active in both root and developing seeds - tissues with high cell division rates where DNA damage is most consequential.

6. sog1-1 Direction Reversal at Sub-Tissue Level

The broad organ analysis showed *sog1-1* reverses the radiation response from up to down. At the sub-tissue level, this reversal is not uniform - some tissues show complete reversal while others maintain the radiation direction:

Sub-tissue	Organ	Radiation dir	Interaction dir	Pattern
seed_stage_9	Seed_Silique	+0.267 (up)	-0.456 (down)	Complete reversal
seed_stage_10	Seed_Silique	+0.285 (up)	-0.447 (down)	Complete reversal
root	Root	+0.131 (up)	-0.321 (down)	Complete reversal
senescing_leaf	Leaf_Shoot	+0.711 (up)	-0.359 (down)	Complete reversal
sepal	Flower	+0.607 (up)	-0.328 (down)	Complete reversal
inflorescence_apex	Leaf_Shoot	+0.013 (neutral)	+0.222 (up)	Same direction (up)
shoot_apex_transition	Leaf_Shoot	-0.020 (neutral)	+0.241 (up)	Same direction (up)

Interpretation

Most sub-tissues show complete direction reversal under *sog1-1*, confirming that *SOG1* is the master regulator of the radiation response across tissues. But shoot apices (inflorescence_apex, shoot_apex_transition) maintain or increase their upregulation in the *sog1-1* interaction. This suggests a *SOG1*-independent radiation response pathway active in meristematic tissues - potentially involving hormonal signaling (auxin/cytokinin) rather than direct DNA damage response.

7. Figures

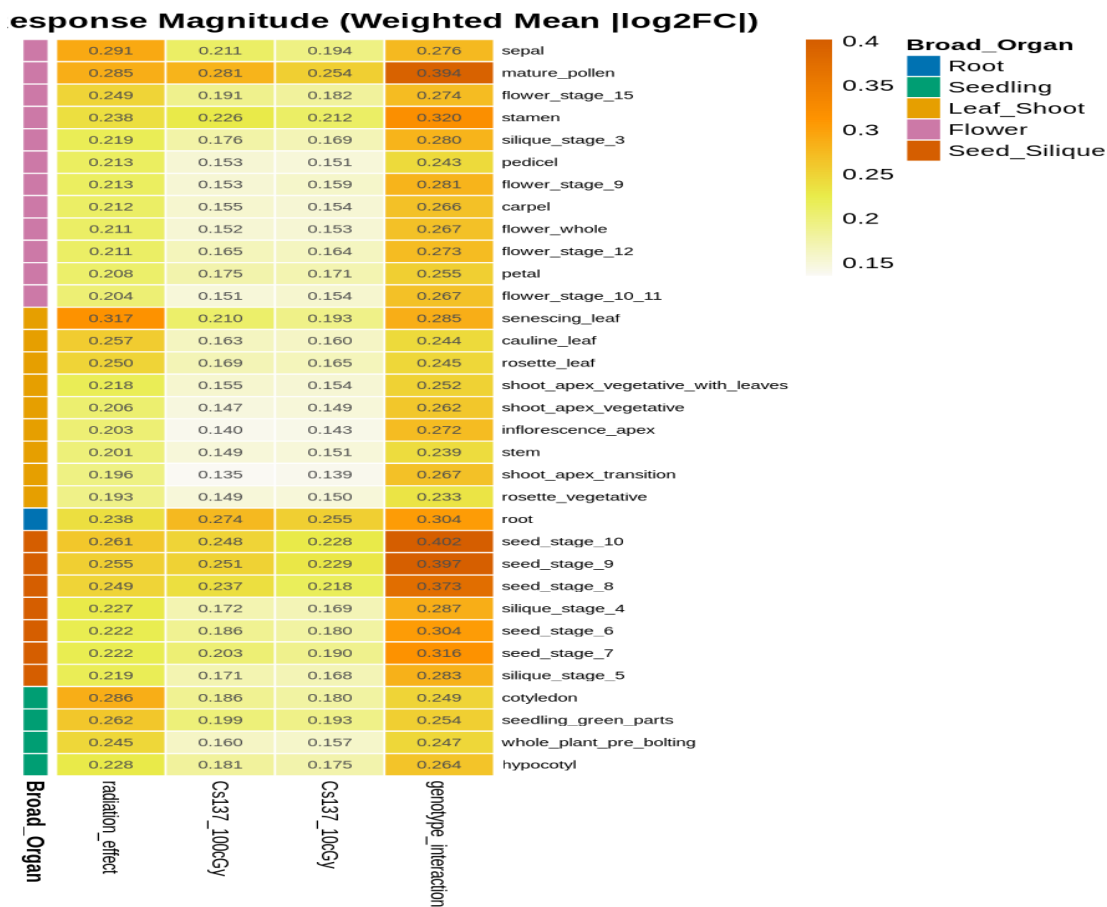


Figure 1: Sub-tissue response magnitude heatmap. Each cell shows the weighted mean |log2FC| for that sub-tissue under each radiation condition. Senescing_leaf, cotyledon, and seedling_green_parts show the highest response magnitudes.

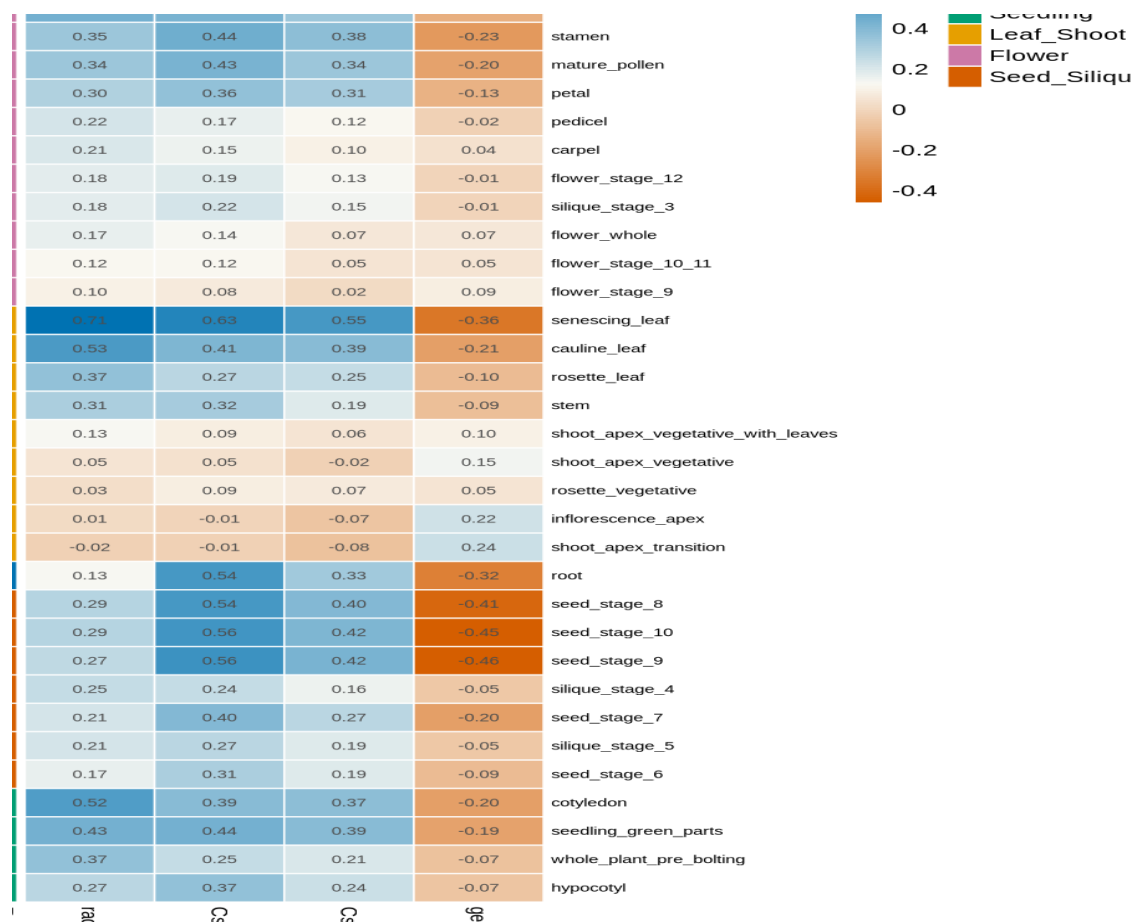


Figure 2: Sub-tissue response direction heatmap. Blue = upregulated, orange = downregulated. The genotype_interaction column shows near-universal reversal compared to radiation_effect.

Tissue-Specific Genes Respond More Strongly to Radiation

Higher Tau = more tissue-specific; positive slope = tissue-specific genes have larger responses

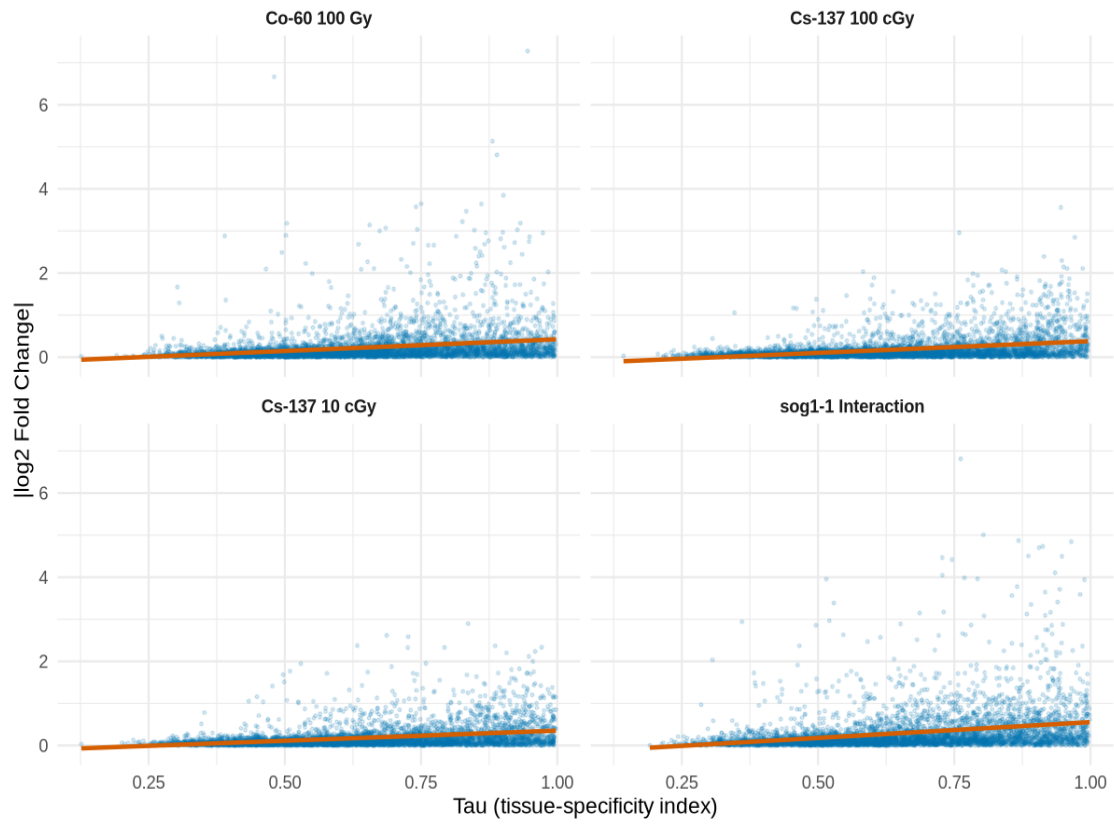


Figure 3: Tau vs $|\log_2 FC|$ scatter. The positive slope in all 4 panels shows that tissue-specific genes (high Tau) have systematically larger radiation responses.

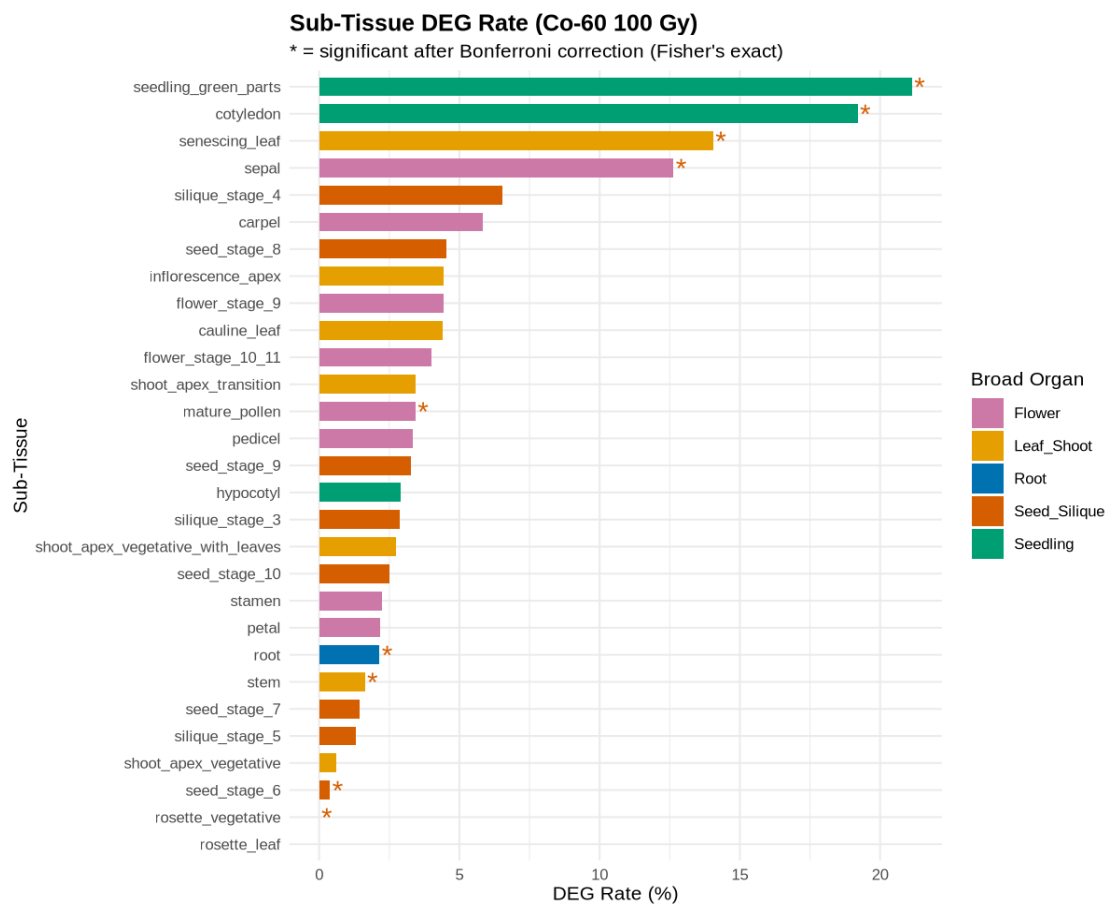


Figure 4: Sub-tissue DEG rate for Co-60 100 Gy. Stars indicate significant enrichment/depletion after Bonferroni correction. Cotyledon and seedling_green_parts have the highest DEG rates (~20%).

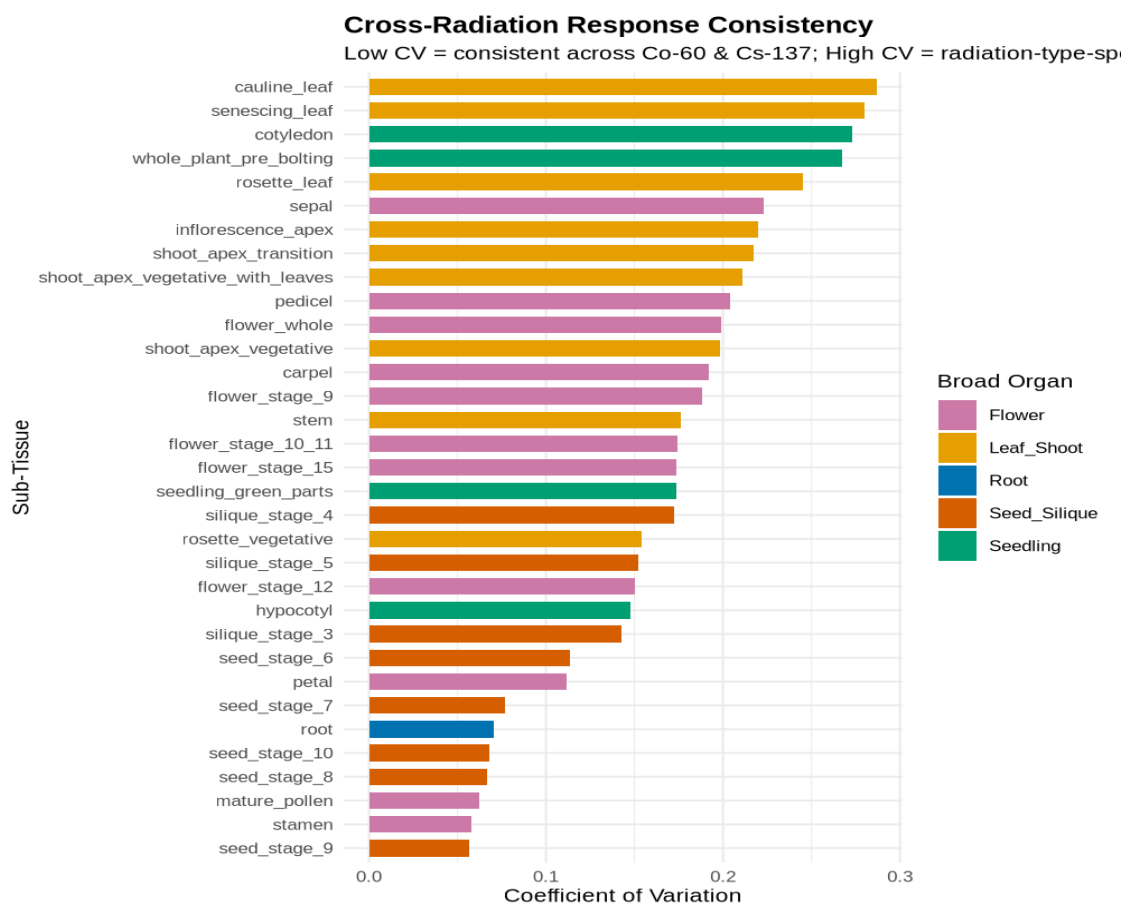


Figure 5: Cross-radiation consistency (coefficient of variation). Seed and pollen tissues are the most consistent across radiation types; photosynthetic tissues are the most radiation-type-specific.

8. Summary of New Findings

1. Sub-tissue enrichment is 10x more informative than organ-level: the Leaf_Shoot enrichment is driven entirely by senescing_leaf (OR=4.0) and cauline_leaf, not by rosette_leaf or shoot apices (which are neutral or depleted).
2. Tissue-specific genes ($\text{Tau} > 0.8$) have 10-25x higher DEG rates than constitutive genes ($\text{Tau} 0.2-0.4$). The radiation response is intrinsically tissue-specific (Spearman $\rho = 0.36-0.41$).
3. Tissue response rankings are conserved across gamma radiation types (Kendall's $W = 0.905$), but the magnitude is radiation-type-specific: Co-60 drives photosynthetic tissue responses, Cs-137 drives root and seed responses.
4. Hierarchical clustering reveals 4 sub-tissue response groups, including a root-seed cluster (Cluster 4) that shares a SOG1-dependent response pattern despite belonging to different organs.
5. sog1-1 direction reversal is nearly universal but not complete: shoot apices maintain upregulation, suggesting a SOG1-independent radiation response in meristematic tissues.
6. Permutation testing confirms tissue-weighted response scores are highly non-random ($Z > 500$ in all comparisons, $p < 0.001$).

Answer to the question

Yes, specific sub-tissue responses can be identified beyond the organ level. The 33 sub-tissue analysis reveals that: (a) cotyledon and seedling_green_parts are the most radiation-responsive tissues (20% DEG rate), not 'Seedling' as a whole; (b) senescing_leaf is the most responsive Leaf_Shoot sub-tissue, while rosette_vegetative has zero DEGs; (c) root and late-stage seeds share a SOG1-dependent response pattern; and (d) the tissue response ranking is conserved across radiation types but the magnitude is radiation-specific. These patterns are invisible at the 5-organ level.